

Queries using JOIN on tables of COMPANY DBMS given in Assgnment 1
(following NAVATHE..Book)

1. List employee name and their department name, using inner-join

Soln:

```
SELECT e.fname, e.lname, d.dname
FROM employee e
INNER JOIN department d
ON e.dno = d.dnumber;
```

2. List all employees and their dependents (if any)
Hint: Use left outer join
3. List all departments and the employees working in them.
Hint: Use right outer join
4. Show all employees and all dependents (matched or unmatched)
Hint: Use full outer join
5. Show employee name and their manager name.Hint: Self join
6. List employee name, project name, and hours worked.Hint: join multiple tables (Employee, Project, Works_on);
7. Show all combinations of employees and projects.
Hint: Show 2 different approaches 1. Using CROSS-JOIN
2. Using table names in FROM clause.

#Simple Aggregate Queries

1. Find the Total Number of Employees

```
SELECT COUNT(*) AS total_employees
FROM employee;
```

2. Average Salary of Employees

```
SELECT AVG(salary) AS avg_salary
FROM employee;
```

3. Maximum Salary of Employee

```
SELECT MAX(salary) AS max_salary
FROM employee;
```

4. Minimum Salary

```
SELECT MIN(salary) AS min_salary
FROM employee;
```

- 5.Total Salary Paid by Company

```
SELECT SUM(salary) AS total_salary
FROM employee;
```

- 6.Number of Employees in Department 5

```
SELECT COUNT(*)  
FROM employee  
WHERE dno = 5;
```

Aggregate Queries with GROUP BY

1. Count Employees in Each Department

```
SELECT dno, COUNT(*) AS total_employees  
FROM employee  
GROUP BY dno;
```

2. Find Average Salary in Each Department.
3. Maximum Salary in Each Department.
4. Total Hours Worked on Each Project.

Using GROUP BY and HAVING

1. Departments Having More Than 3 Employees

```
SELECT dno, COUNT(*) AS total_employees  
FROM employee  
GROUP BY dno  
HAVING COUNT(*) > 3;
```

2. List Departments with Dnos, with Average Salary Greater Than 40000

#Use of GROUPBY clause with JOIN

8. Count Employees in Each Department

```
SELECT d.dname, COUNT(e.ssn) AS total_employees  
FROM department d  
JOIN employee e  
ON d.dnumber = e.dno  
GROUP BY d.dname;
```

9. List the number of Employees Working on Each Project.

10. Departments Having More Than 3 Employees